A Seminar Report

On

"MULTIMODAL RETRIEVAL-AUGMENTED GENERATION"

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CERTIFICATE

This is to certify that the seminar report entitled "Smart Tree Adoption System: An AIEnabled Platform for Sustainable Environmental Conservation" being submitted by Adhiraj Sikandar Kate record of bonafide work carried out by him under the supervision and guidance of prof. A.S Thorat in partial fulfillment of the requirement for **TE (Computer Engineering)** 2024 course of Savitribai Phule Pune University,Pune in the academic year 2026-27

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Abstract

Multimodal Retrieval-Augmented Generation (Multimodal RAG) is an emerging approach that combines information retrieval with generative artificial intelligence to enable Large Language Models (LLMs) to retrieve and reason over information from multiple modalities. Traditional Retrieval-Augmented Generation systems primarily retrieve textual information from external knowledge sources, which can limit their ability to understand visually rich and heterogeneous documents containing images, tables, charts, diagrams, and other forms of data. Multimodal RAG addresses this limitation by integrating multimodal retrieval, embedding techniques, document processing, context fusion, and multimodal language models.

In a Multimodal RAG system, information from different sources is processed and represented in a form suitable for retrieval. Relevant textual and visual information is then retrieved based on the user query and provided as contextual evidence to a multimodal generative model. This enables the system to generate responses that incorporate information from multiple modalities. Recent research has explored multimodal document understanding, visually rich document retrieval, adaptive retrieval, image retrieval, retrieval-reflection mechanisms, and evaluation benchmarks.

This seminar presents a systematic study of Multimodal RAG, including its architecture, working methodology, retrieval and generation processes, recent research developments, applications, advantages, and limitations. A comparative analysis of recent research approaches is also presented based on their methodologies, supported modalities, retrieval techniques, evaluation approaches, and research contributions. The seminar further discusses challenges such as cross-modal alignment, retrieval accuracy, context integration, hallucination, computational cost, privacy, security, and reliable evaluation. Finally, potential future research directions including agentic multimodal retrieval, multilingual systems, improved multimodal embeddings, privacy-preserving approaches, and better evaluation frameworks are discussed.

Keywords: Multimodal Retrieval-Augmented Generation, Retrieval-Augmented Generation, Large Language Models, Multimodal AI, Multimodal Large Language Models, Vision-Language Models, Multimodal Retrieval, Information Retrieval, Embeddings, Generative AI.

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List of Symbols and Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
RAG	Retrieval-Augmented Generation
MRAG	Multimodal Retrieval-Augmented Generation
LLM	Large Language Model
MLLM	Multimodal Large Language Model
VLM	Vision-Language Model
NLP	Natural Language Processing
OCR	Optical Character Recognition
VQA	Visual Question Answering
QA	Question Answering
API	Application Programming Interface
PDF	Portable Document Format
VDB	Vector Database
ANN	Approximate Nearest Neighbor
CLIP	Contrastive Language–Image Pre-training
FAISS	Facebook AI Similarity Search

Chapter 1: Introduction

1.1 Overview of Multimodal Retrieval-Augmented Generation

The rapid development of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Generative AI has significantly changed the way information is processed and accessed. Large Language Models (LLMs) have demonstrated strong capabilities in understanding natural language and generating human-like responses. However, LLMs mainly depend on the information available during their training and may produce inaccurate or outdated responses when they do not have access to relevant external knowledge.

Retrieval-Augmented Generation (RAG) was introduced to address this limitation by combining information retrieval with text generation. Instead of relying only on the knowledge stored within a language model, a RAG system retrieves relevant information from an external knowledge source and provides it to the language model as additional context. This approach can improve the relevance and factual grounding of generated responses.

Traditional RAG systems are primarily designed to work with textual information. However, real-world information is often available in multiple forms, including text, images, tables, charts, diagrams, scanned documents, and other visual content. Important information may be lost when such content is converted into plain text.

Multimodal Retrieval-Augmented Generation (Multimodal RAG) extends the traditional RAG approach by enabling retrieval and processing of information from multiple modalities. It can combine textual and visual information during retrieval and generation, allowing multimodal models to use a broader range of evidence when producing responses.

A typical Multimodal RAG system consists of several stages, including multimodal data collection, preprocessing, embedding generation, information retrieval, context integration, and response generation. Depending on the system architecture, different modalities can be represented separately or within a shared embedding space. The retrieved information is then provided to a multimodal generative model for generating a response.

Multimodal RAG has potential applications in areas such as intelligent document processing, education, healthcare information systems, visual question answering, enterprise knowledge management, technical document analysis, and digital libraries. Recent research has also focused on improving retrieval accuracy, multimodal reasoning, document understanding, adaptive retrieval, and evaluation methods. Therefore, Multimodal RAG represents an important research direction at the intersection of information retrieval, multimodal artificial intelligence, computer vision, natural language processing, and generative AI.

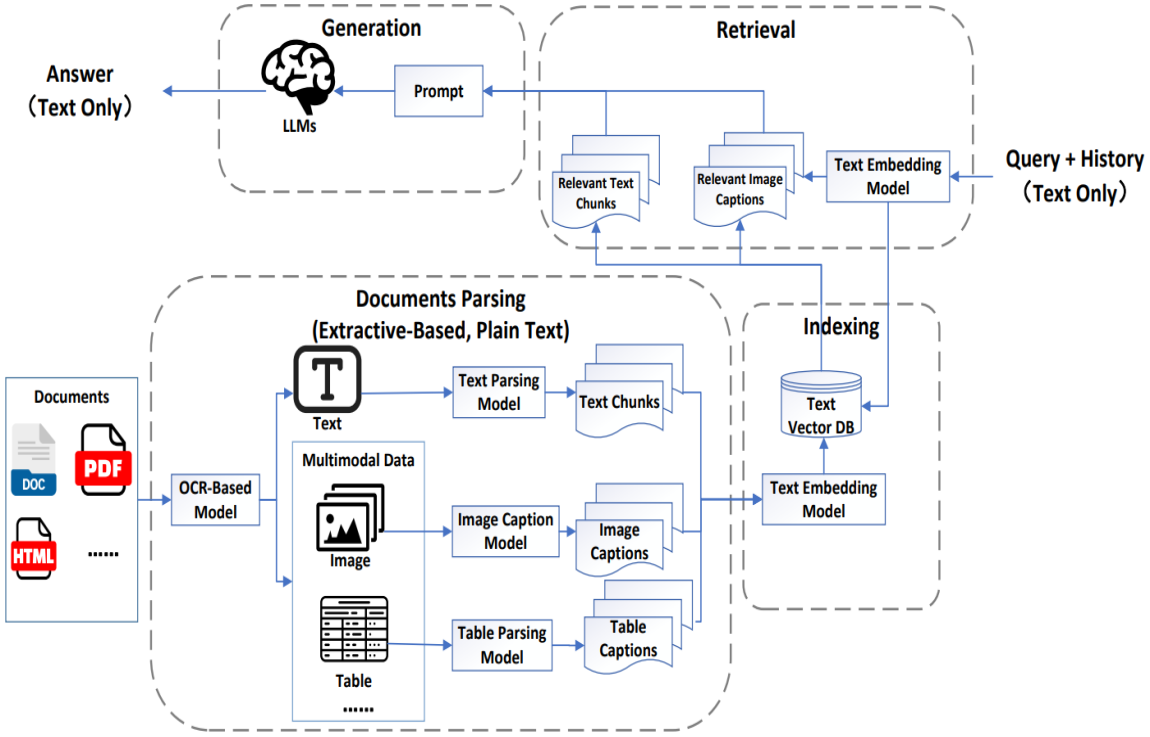


Figure 1.1: Overview of Multimodal Retrieval-Augmented Generation

1.2 Background of Generative AI and Large Language Models

Artificial Intelligence refers to the development of systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, perception, language understanding, and decision-making.

Machine Learning is a subfield of Artificial Intelligence in which systems learn patterns from data and use those patterns to make predictions or decisions. Deep Learning further extends Machine Learning through multi-layered neural networks that can process complex data such as images, speech, and text. Generative AI is a branch of Artificial Intelligence that focuses on creating new content based on learned patterns. Depending on the model, the generated content can include text, images, audio, video, or other forms of data.

Large Language Models are deep-learning models trained on large collections of textual data. They can perform tasks such as text generation, summarization, question answering, translation, and information extraction. However, LLMs may face limitations when dealing with information that is not represented effectively in their input or training data.

The development of Multimodal Large Language Models (MLLMs) and Vision-Language Models (VLMs) has expanded AI systems beyond text-only processing. These models can process combinations of textual and visual information, making them suitable for applications requiring understanding of images, documents, charts, and other visual elements.

1.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is a technique that combines a retrieval system with a generative model. The main objective of RAG is to provide a generative model with relevant external information before generating an answer.
In a traditional RAG system, a user's query is first converted into a representation suitable for searching a knowledge base. Relevant documents or document chunks are then retrieved using similarity-based search. These retrieved results are provided to the language model as context, and the model generates the final response using both the query and retrieved information.

1.3.1 Retrieval

Retrieval is the process of finding relevant information from an external knowledge source. Documents are generally divided into smaller chunks and converted into numerical representations called embeddings.

When a user submits a query, the query is also converted into an embedding. The system then compares the query representation with stored document representations to identify relevant information.

1.3.2 Generation

After retrieving relevant information, the retrieved content is provided to a generative model as contextual information. The model uses this context along with the user's query to generate an appropriate response.

1.3.3 Knowledge Base and Vector Database

A RAG system can use an external knowledge base containing documents, web pages, reports, manuals, or other information sources. A Vector Database (VDB) can be used to store embeddings and perform similarity searches efficiently.

1.3.4 Complete RAG Workflow

The general workflow of traditional RAG can be represented as:
User Query → Query Processing → Query Embedding → Vector Database → Relevant Documents → Context Building → LLM → Generated Answer

1.3.5 Traditional RAG Architecture

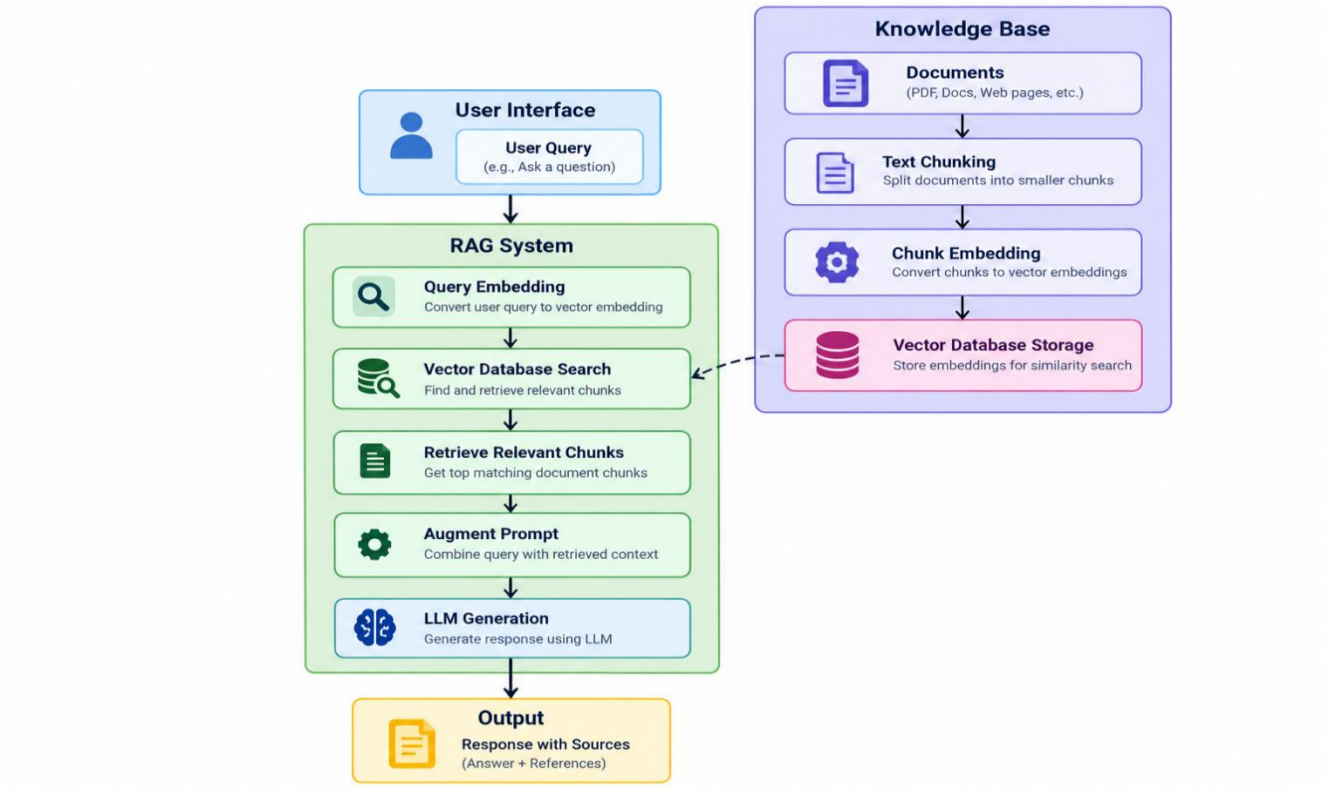


Figure 1.2: Architecture of Traditional Retrieval-Augmented Generation

1.3.6 Advantages of RAG

RAG provides several advantages over generation based only on a model's internal knowledge. It can incorporate information from external knowledge sources, improve the relevance of responses, and support knowledge that can be updated independently of the model.

1.3.7 Limitations of Traditional RAG

Traditional RAG systems generally focus on textual information. Consequently, they may have difficulty effectively representing and retrieving information contained in images, charts, tables, diagrams, and visually rich documents. Converting these elements into plain text can also result in the loss of layout and visual relationships.

These limitations provide motivation for extending RAG systems to support multiple modalities.

1.4 Limitations of Traditional Retrieval-Augmented Generation

Although RAG improves the use of external knowledge, traditional text-oriented RAG systems have several limitations when dealing with multimodal information.

1.4.1 Primarily Text-Oriented Retrieval

Many traditional RAG systems are designed mainly for text documents. When important information exists in visual form, text-based retrieval may not be sufficient to retrieve the required evidence.

1.4.2 Difficulty in Understanding Images

Images may contain information that cannot be completely represented through surrounding text. For example, photographs, diagrams, medical images, and technical illustrations may contain important visual information.

1.4.3 Difficulty in Processing Charts and Graphs

Charts and graphs communicate relationships through visual structures such as axes, labels, lines, and bars. Converting them into plain text can remove some of these relationships and make retrieval more difficult.

1.4.4 Difficulty in Understanding Tables

Tables contain structured relationships between rows and columns. Simple text extraction may disturb this structure, making it difficult for a retrieval system or language model to correctly understand the relationship between values.

1.4.5 Loss of Document Layout Information

Visually rich documents such as reports, research papers, invoices, and presentations use layout to communicate relationships between different elements. Text-only processing may lose information about the position and relationship of these elements.

1.4.6 OCR Errors

Scanned documents and images often require Optical Character Recognition (OCR) to extract text. OCR errors can introduce incorrect characters or words, which may negatively affect subsequent retrieval and generation.

1.4.7 Context Fragmentation

When multimodal documents are converted into independent text chunks, relationships between text, images, tables, and their surrounding content may be lost. This can result in incomplete context being provided to the generative model.

1.4.8 Retrieval Errors

If the retrieval system fails to identify the correct document or relevant information, the generative model may receive insufficient or irrelevant context.

1.4.9 Hallucination

A generative model can sometimes produce information that is not supported by the retrieved evidence. Effective retrieval and grounding are therefore important for improving response reliability.

1.4.10 Limited Cross-Modal Reasoning

Text-only RAG has limited ability to reason across different types of information. For example, answering a question may require combining a paragraph with a chart or an image with its accompanying text.

1.4.11 Summary of Limitations

Table 1.1: **Limitations of Traditional RAG**

Sr. No.	Limitation	Impact
1	Text-oriented retrieval	Difficulty retrieving visual information
2	Image understanding	Important visual details may be missed
3	Chart processing	Visual relationships may be lost
4	Table processing	Row-column relationships may be affected
5	Layout loss	Document structure may not be preserved
6	OCR errors	Extracted information may become inaccurate
7	Context fragmentation	Relationships between modalities may be lost
8	Retrieval errors	Irrelevant context may be supplied
9	Hallucination	Responses may not be fully grounded
10	Limited cross-modal reasoning	Difficult to combine text and visual evidence

1.5 Need for Multimodal Retrieval-Augmented Generation

The increasing availability of multimodal data has created a need for systems that can retrieve and reason over information beyond plain text. Modern documents frequently contain a combination of paragraphs, images, tables, charts, diagrams, and other visual elements.

Multimodal RAG addresses this requirement by extending the retrieval process to multiple information modalities. Instead of retrieving only text passages, the system can retrieve relevant combinations of textual and visual information and provide them to a multimodal generative model.
For example, if a user asks a question about a research report containing a graph, a traditional text-based RAG system may retrieve the paragraph describing the graph. A Multimodal RAG system can potentially retrieve both the relevant paragraph and the graph itself, allowing the multimodal model to use both sources of information.

Chapter 2: Literature Survey

2.1 Introduction

Multimodal Retrieval-Augmented Generation (Multimodal RAG) is an emerging approach that combines information retrieval with multimodal generative models. Recent research focuses on retrieving and processing information from text, images, tables, charts, and visually rich documents.

This chapter reviews selected recent research works related to Multimodal RAG and identifies the major approaches, contributions, and research gaps.

2.2 Review of Recent Research

2.2.1 Multimodal RAG Evaluation Benchmark

This research focuses on evaluating Multimodal RAG systems. It highlights the need for suitable benchmarks to measure retrieval quality, multimodal understanding, and generation performance.

2.2.2 VDocRAG

VDocRAG focuses on Retrieval-Augmented Generation over visually rich documents. It addresses the limitations of text-only processing by considering visual and structural information present in documents.

2.2.3 SAM-RAG

Self-Adaptive Multimodal RAG introduces an adaptive retrieval approach. It aims to select relevant information according to the requirements of different queries and modalities.

2.2.4 mR²AG

mR²AG combines multimodal retrieval with a reflection mechanism for knowledge-based Visual Question Answering. The approach focuses on improving the use of retrieved evidence during multimodal reasoning.

2.2.5 Multimodal RAG Survey

The survey **"Ask in Any Modality"** provides an overview of Multimodal RAG research, including multimodal retrieval, representations, generation, evaluation, challenges, and future directions.

2.2.6 DeepSem

DeepSem investigates semantic image retrieval for Multimodal RAG, with a focus on improving the retrieval of relevant visual information for industrial applications.

2.3 Comparative Analysis

Table 2.1: Comparative Analysis of Selected Research

Research Work	Main Focus	Major Contribution
Multimodal RAG Evaluation Benchmark	Evaluation	Benchmark for Multimodal RAG
VDocRAG	Visual documents	Visual and structural document understanding
SAM-RAG	Adaptive retrieval	Query-dependent multimodal retrieval
mR²AG	Visual Question Answering	Retrieval with reflection
Ask in Any Modality	Survey	Overview of Multimodal RAG research
DeepSem	Image retrieval	Semantic visual retrieval

2.4 Research Gap

The literature shows that Multimodal RAG has advanced significantly, but several challenges remain. These include **cross-modal alignment, retrieval accuracy, context integration, hallucination, computational cost, and reliable evaluation**. Further research is required to develop efficient systems that can accurately retrieve and combine information from multiple modalities while maintaining reliable and grounded responses.

2.5 Summary

The reviewed research demonstrates that Multimodal RAG is developing across areas such as visual document understanding, adaptive retrieval, multimodal reasoning, image retrieval, and system evaluation. These studies provide the foundation for understanding the architecture and methodology of Multimodal RAG discussed in the next chapter.

Chapter 3: Software Requirement Specification

3.1 Introduction

This chapter describes the software and technical requirements for studying and developing a basic prototype of a Multimodal Retrieval-Augmented Generation (Multimodal RAG) system. The requirements include the hardware, software, functional, and non-functional requirements necessary for implementing the system.

3.2 Hardware Requirements

The basic hardware requirements are:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 20 GB available space
- **GPU:** Optional; recommended for running large multimodal models locally
- **Internet:** Required for accessing APIs, models, and online resources

3.3 Software Requirements

The software requirements include:

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **Development Environment:** Visual Studio Code / Jupyter Notebook
- **Libraries:** PyTorch, Transformers, NumPy, Pandas
- **Document Processing:** PyMuPDF / similar PDF processing library
- **Vector Database:** FAISS / ChromaDB
- **Embedding Models:** Text and multimodal embedding models
- **Generative Model:** Large Language Model or Multimodal Large Language Model
- **Version Control:** Git and GitHub

3.4 Functional Requirements

The system should be capable of performing the following functions:

1. Accept user queries in text or supported multimodal formats.
2. Process documents containing text and visual information.
3. Generate suitable embeddings for the available information.
4. Store embeddings in a vector database.
5. Retrieve relevant information based on the user query.
6. Combine retrieved information from different modalities.
7. Provide the retrieved context to a generative model.
8. Generate a relevant response based on the retrieved information.

3.5 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- **Accuracy:** Retrieved information and generated responses should be relevant.
- **Performance:** Retrieval and response generation should be completed within reasonable time.
- **Scalability:** The system should support increasing amounts of multimodal data.
- **Reliability:** The system should provide consistent results for similar queries.
- **Security:** User data and stored documents should be protected.
- **Usability:** The system should have a simple and understandable interface.
- **Maintainability:** Components should be modular and easy to update.

3.6 Summary

The Software Requirement Specification defines the basic requirements for a Multimodal RAG system. These requirements provide a foundation for designing the system architecture and methodology discussed in the next chapter.

Chapter 4: System Architecture

4.1 Introduction

The system architecture of Multimodal Retrieval-Augmented Generation (Multimodal RAG) describes the flow of information from data input to final response generation. The system integrates multimodal data processing, embedding generation, information retrieval, context fusion, and a multimodal generative model.

4.2 Multimodal RAG Architecture

The major components of the architecture are:

1. Multimodal Data Sources
2. Data Preprocessing
3. Embedding Generation
4. Multimodal Retrieval
5. Context Fusion
6. Multimodal Language Model
7. Generated Response

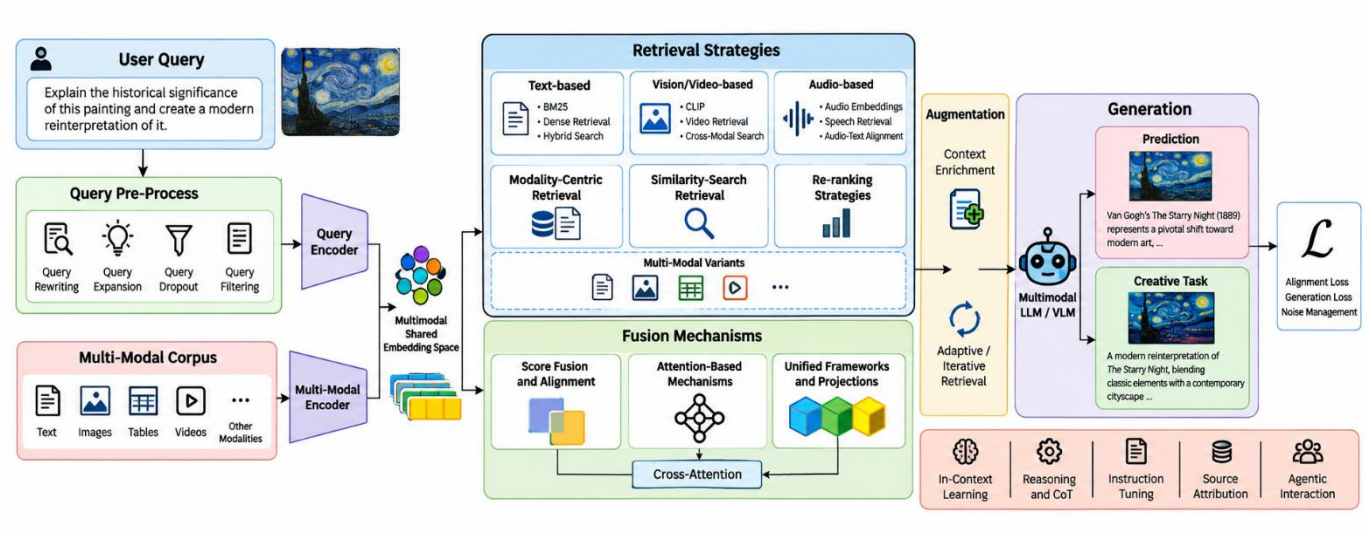


Figure 4.1: Architecture of Multimodal Retrieval-Augmented Generation

4.3 Data Ingestion and Preprocessing

The first stage involves collecting information from different sources such as text documents, PDFs, images, tables, and charts. The collected data is processed to make it suitable for embedding and retrieval.

Text can be extracted from documents, while images and visual elements can be processed using suitable computer vision or multimodal processing techniques.

Chapter 5: Advantages & Disadvantages

5.1 Introduction

Multimodal Retrieval-Augmented Generation (Multimodal RAG) extends traditional RAG by allowing systems to retrieve and use information from multiple modalities such as text, images, tables, charts, and documents. This provides several benefits for information retrieval and generation, but it also introduces technical and practical challenges.

5.2 Advantages

5.2.1 Supports Multiple Modalities

Multimodal RAG can process information from different sources, including text, images, tables, charts, and visually rich documents. This allows the system to work with more types of real-world information.

5.2.2 Improved Contextual Understanding

By combining information from multiple modalities, the system can obtain richer context for answering user queries.

5.2.3 Better Document Understanding

Multimodal RAG can preserve and utilize visual and structural information from documents that may be lost in text-only processing.

5.2.4 More Relevant Responses

Retrieving relevant information before generation can help the model produce responses that are better aligned with the available external knowledge.

5.2.5 Supports Cross-Modal Reasoning

The system can combine information from different modalities, such as using text together with an image, table, or chart to answer a query.

5.2.6 Useful for Multiple Applications

Multimodal RAG can be applied to areas such as education, healthcare, intelligent document processing, technical documentation, visual question answering, and enterprise knowledge management.

5.3 Disadvantages

5.3.1 High Computational Cost

Processing multiple modalities requires additional computational resources for data processing, embedding generation, retrieval, and response generation.

5.3.2 Complex System Architecture

Multimodal RAG involves several components, including preprocessing modules, embedding models, vector databases, retrieval mechanisms, and multimodal generative models, making the system more complex than traditional RAG.

5.3.3 Retrieval Errors

If the system retrieves irrelevant or incomplete information, the quality of the generated response may be affected.

5.3.4 Cross-Modal Alignment Challenges

Different modalities have different representations. Correctly connecting related text, images, tables, and other information can be challenging.

5.3.5 Hallucination

Even with retrieved information, the generative model may produce information that is not supported by the available evidence.

5.3.6 Data and Model Dependency

The performance of a Multimodal RAG system depends on the quality of the input data, embedding models, retrieval methods, and multimodal generative models.

5.3.7 Privacy and Security Concerns

Multimodal systems may process sensitive documents, images, and other information. Proper data protection and access control are therefore required.

5.4 Applications

Multimodal Retrieval-Augmented Generation (Multimodal RAG) can be applied in domains where information is available in multiple forms such as text, images, tables, charts, and documents. By retrieving and combining relevant information from different modalities, Multimodal RAG can support more comprehensive information processing and response generation.

Chapter 6: Conclusion & Future Scope.

6.1 Conclusion

Multimodal Retrieval-Augmented Generation (Multimodal RAG) is an emerging approach that combines information retrieval with multimodal generative models. Unlike traditional RAG systems that mainly work with text, Multimodal RAG can utilize information from different modalities such as text, images, tables, charts, and visually rich documents.

The literature survey shows that recent research has focused on improving multimodal retrieval, visual document understanding, adaptive retrieval, multimodal reasoning, image retrieval, and evaluation. By retrieving relevant information and providing it as context to a multimodal model, Multimodal RAG can support more comprehensive information processing and response generation.

However, challenges such as retrieval accuracy, cross-modal alignment, context integration, computational cost, hallucination, privacy, and reliable evaluation still need to be addressed. Therefore, Multimodal RAG remains an active research area with significant opportunities for further development.

6.2 Future Scope

The future scope of Multimodal RAG includes:

1. **Improved Multimodal Embeddings:** Development of better embeddings for accurately representing relationships between text, images, tables, and other modalities.
2. **Advanced Retrieval Techniques:** Development of more accurate and efficient retrieval methods for selecting relevant multimodal information.
3. **Agentic Multimodal RAG:** Integration of AI agents to dynamically decide what information to retrieve and how to process it.
4. **Multilingual Multimodal RAG:** Development of systems capable of handling multiple languages along with visual information.
5. **Reduced Hallucination:** Development of improved grounding and verification techniques to generate more reliable responses.
6. **Efficient Processing:** Optimization of models and retrieval systems to reduce computational cost and response time.
7. **Privacy-Preserving Systems:** Development of techniques that protect sensitive multimodal data while enabling effective retrieval and generation.
8. **Improved Evaluation:** Development of standardized benchmarks and evaluation metrics for measuring multimodal retrieval, reasoning, and generation performance.

Overall, future research can focus on making Multimodal RAG systems **more accurate, efficient, reliable, secure, and capable of handling diverse real-world multimodal information**.

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